# # IUPAC NOMENCIATURE - Organic Chemistry

→ Basic Terms of Organic Chemistry.

- Hydrocarbon :- Only hydrogen & Carbon Containing Compounds.

  eg → $C_6H_6$, $C_2H_4$, $C_2H_2$ etc.

- Organic Compound :- Hydrocarbons & derivaties of Hydrocarbon are called organic Compound.

  ⇒ Tetravalent Carbon (essential)

  ⇒ H → F, Cl, Br, I (Halogen), N, O, S, P etc

  ⇒ Covalent bond Containing Compounds.

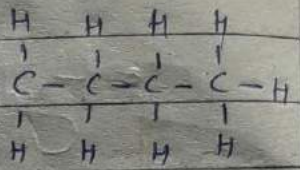
- Catenation :- long chain formation tendency of same atoms.

  ⇒ max$^{n}$ in Carbon due to tetravalency & due to strong bond formation

  C-C > Si-Si > P-P > S-S

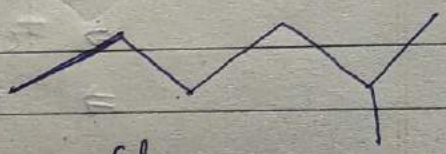
- Molecular formula :- It indicate no. of atom of each element present in one molecule.

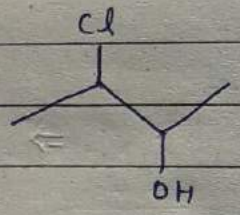
- Structural formula :- It indicates connectivity of atoms in one molecule.

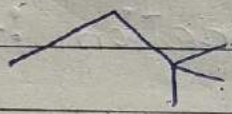
→ Expanded str. formula → 

→ Condensed str. formula → $CH_3-CH_2-CH_3$
or $CH_3(CH_2)CH_3$

→ Bond line notation → 

eg → $CH_3(CH_2)_3CH(CH_3)_2$ → 

eg → $CH_3CHClCH(OH)CH_3$ → 


eg → $CH_3CH_2C(CH_3)_3$ → 

eg → $C(CH_3)_4$ → 

- Homologous Series :- Series of Compound which have

→ Same general formula
→ Same functional group.
→ Same chemical properties

→ difference between molecular formula of two member of this series is equal to $n \times CH_2$.

→ Same degree of Unsaturation

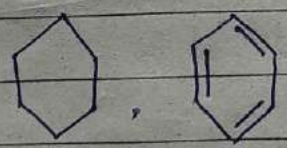
→ different Emperical formula.
→ different molecular formula.
→ different physical property.

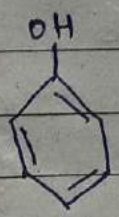
| eg→ Alkane | Alkene | Alkyne | Alkohol | Car. acid |
|---|---|---|---|---|
| $C_nH_{2n+2}$ | $C_nH_{2n}$ | $C_nH_{2n-2}$ | $C_nH_{2n+1}OH$ | $C_nH_{2n}O_2$ or $C_nH_{2n+1}COOH$ |
| $CH_4$ | $C_2H_4$ | $C_2H_2$ | $CH_3-OH$ | $CH_2O_2$ |
| $C_2H_6$ | $C_3H_6$ | $C_3H_4$ | $C_2H_5-OH$ | $C_2H_6O_2$ |

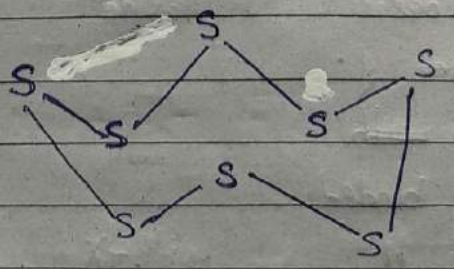
3. • Homocyclic & Heterocyclic Compounds :-

↓
Cyclic Compound Containing identical atoms in ring

eg:-

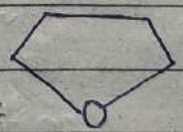




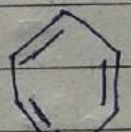


↓
Cyclic Compounds in which atleast one heteroatom (O, S, N, P etc) present in ring.

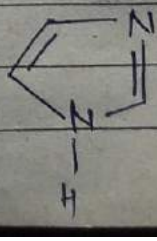
eg→



THF



Pyridine



Date :

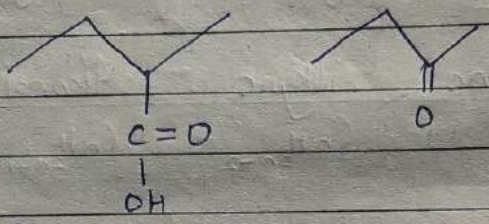
## • Saturated & Unsaturated Compounds :-

- Saturated → only c-c single bond containing Compounds
- Unsaturated → Carbon-Carbon multiple bond (c=c, c≡c) (atleast one) containing Compounds

5.

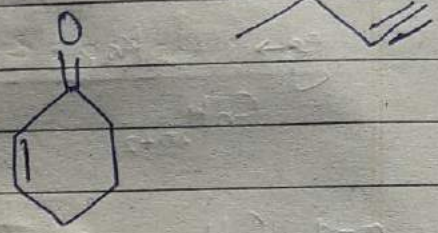
eg.






eg:-






* ~~Identification of bonds~~

4.

• Identification of degree

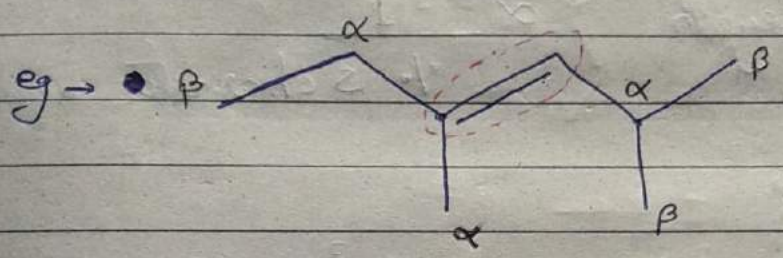
- 1° Primary
- 2° Secondary
- 3° Tertiary
- 4° Quaternary

- for Carbon → on the basis of no. of bonded Carbon atoms
- for Hydrogen → on the basis of degree of Carbon which bonded with H
- for Alkyl Halide → with -X
- for alcohols → with -OH
- for amine → on the basis of no. of bonded Carbon atoms with -N
- Amide → with N

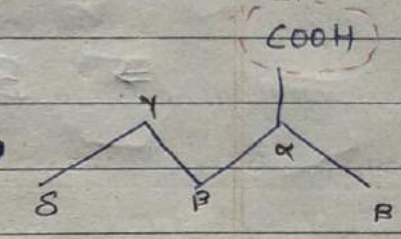
Note :- At Primary Carbon → 3 hydrogen
" Secondary Carbon → 2 "
" Tertiary Carbon → 1 "
" Quarternary Carbon → 0 "

5. • Identification of Relative position w.r.t functional group. { α, β, γ, δ, ε ..... }

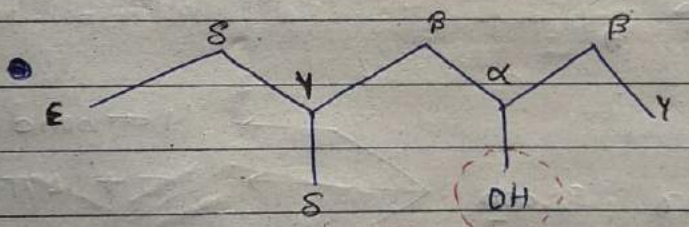
eg →



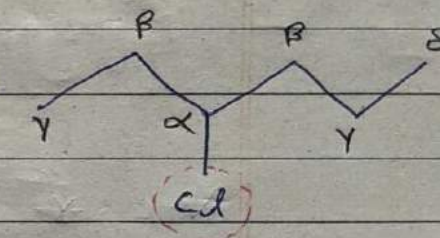

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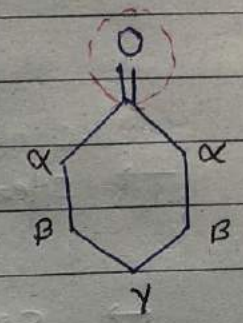

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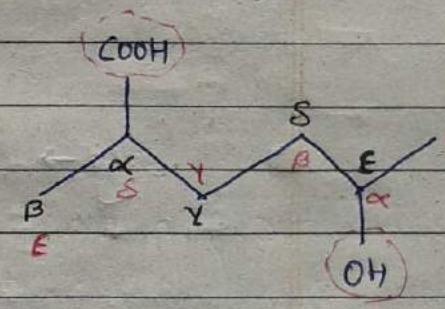

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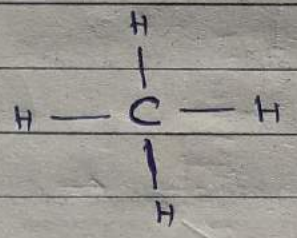

• Identification of hybridization :-

Steric N.O = no. of σ bonds + no. of lone pairs
(non-bonded e⁻ pair)

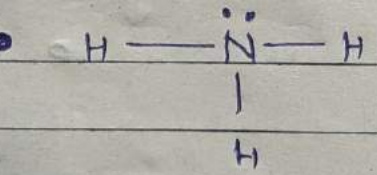
| Steric no. | Hybridisation | % s character |
|---|---|---|
| 2 | SP | 50 % |
| 3 | $SP^2$ | 33.33 % |
| 4 | $SP^3$ | 25 % |

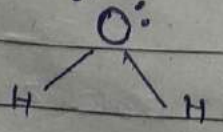
⇒ Electronegativity $\propto$ % s character.

⇒ Bond length $\propto \frac{1}{\%\ s\ character}$.

eg →  → $SP^3$

Note * for Carbon →
- 4σ bond → $SP^3$
- 3σ + 1π → $SP^2$
- 2σ + 2π → SP

eg → 
→ Steric No = 3 + 1 = 4
→ $SP^3$

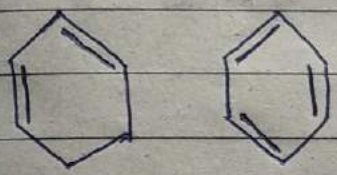
• 
→ Steric no = 2 + 2 = 4
→ $SP^3$

6.

- **Conjugative** → Alternate position/ arrangement of double bonds.

eg→ 





- **Cumulated** → Continuous double bonds

eg→ $CH_2=C=CH_2$

$CH_2=C=C=CH_2$

- **Isolated** → Neither Conjugated Nor - Cumulated.

g:→ 





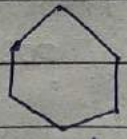
- **Double bonds**

- **Aliphatic** → All non-aromatic Compounds (cyclic or acyclic)

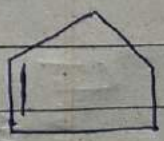
eg→ 

- acyclic
- Aliphatic
- non-aromatic

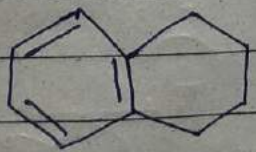
- **Alicyclic** → Aliphatic cyclic or Non-aromatic Cyclic Compound.



- Aliphatic or Alicyclic



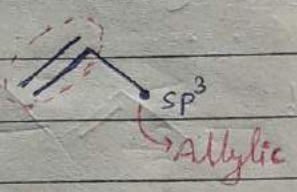
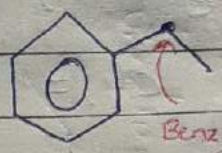
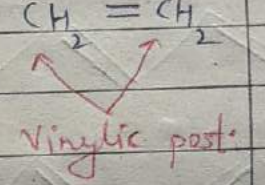
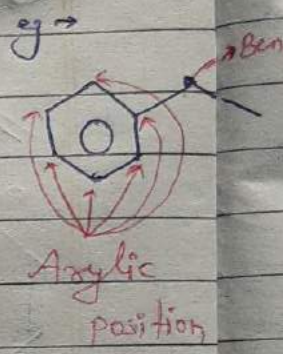
- Aliphatic or Alicyclic



→ Aromatic



→ Aliphatic or Alicyclic

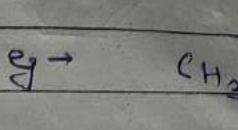
| Allylic | Benzylic | Vinylic | Acylic |
|---|---|---|---|
| ↓ | ↓ | ↓ | ↓ |
| Adjacent $sp^3$ Carbon of C=C of alkene | $sp^3$ Carbon of adjacent position of aromatic ring | $sp^2$ position of (C=C) double bond of alkene. | $sp^2$ position aromatic ring |
| eg →   | eg →   | eg →   | eg →   |

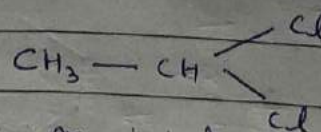
Propargylic → Adjacent $sp^3$ Carbon of C≡C of alkyne

eg -  

- Geminal & Vicinal Compounds.

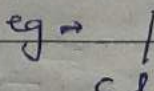
Geminal ↓ when two same group are present one same position Carbon.

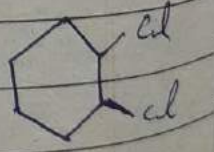
eg → 

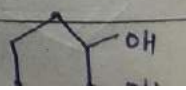


gem. dichalide

Vicinal ↓ when two same groups are present on adjacent position.

eg → 





7. • Degree of Unsaturation (D.U.) / Double bond equiv. (DBE)

or Hydrogen deficiency index :-

Degree of unsaturation is calculation that determines total numbers of π bonds & rings in given Compound.

benzylic

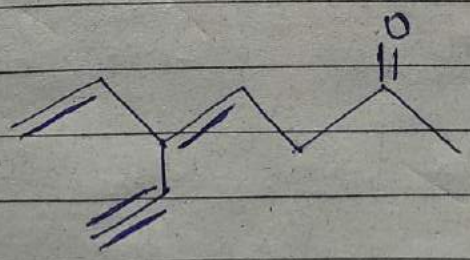
It is deficiency of even no. of hydrogen atoms with respect to most saturated related Compound.

$$DU = \text{no. of } \pi \text{ bonds} + \text{no. of rings}$$

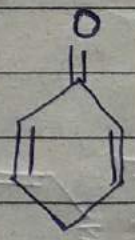
$$DU = \text{no. of Carbon atom} - \left[\frac{H + X - N - 2}{2}\right]$$

Note :- # Same general formula → same DU.

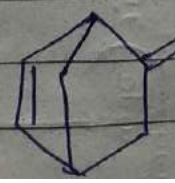
8. eg →



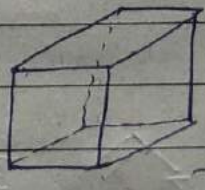

DU = 5




DU = 5




DU = 4

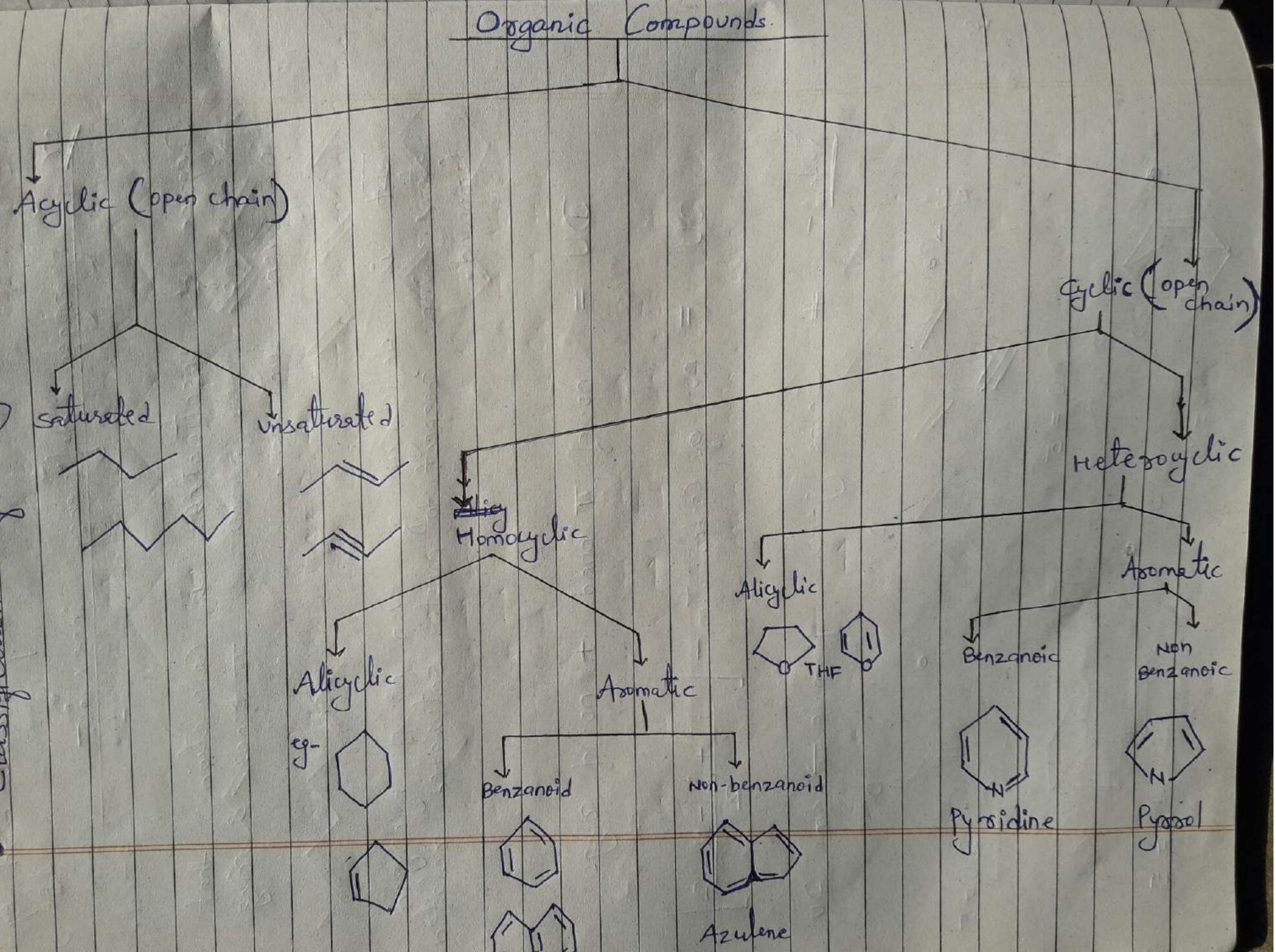


$C_8H_8$

DU = 5

10/

# Classification of Organic Compounds



→ Some important Priority order for naming :-

I. Priority order for main (Parent) chain selection :-

a. Max. no. of functional group.
b. Max no of C-C multiple bonds.
c. Max no of - Carbon atom in main chain
d. Max no of substituents / side chain
e. Lowest set of position (Locant) of F.G > C-C multiple bonds > substituent.
f. Alphabetical order of name of substituents.

II. Priority orders for numbering of main chain :-

a. Lowest set of position of functional group.
b. Lowest set of position of C-C multiple bond.
c. Lowest set of position of substituents
d. Alphabetical orders of name of substituents.

III. Sequence for writing of IUPAC name :-

2° Prefix + 1° Prefix + Word Root + 1° suffix + 2° Suffix
(a) (b) (c) (d) (e)
(ALK)

a. 2° Prefix → name of substituent in alphabetical order (with position)

b. 1° Prefix → Useful for cyclic Compounds.

eg → Cyclo / bicyclo / tricyclo / spiro / bispiro etc.

c. Word root (ALK) :→ It indicates no. of Carbon atoms in main chain.

| | |
|---|---|
| $C_{10}$ → Dec | $C_{13}$ → tridec |
| $C_{11}$ → Undec | $C_{18}$ → Octadec |
| $C_{12}$ → Dodec | $C_{19}$ → Nonadec |
| | $C_{20}$ → Isoc |

d. 1° suffix → It indicates multiplicity of Carbon - Carbon bonds in main chain. (with position).

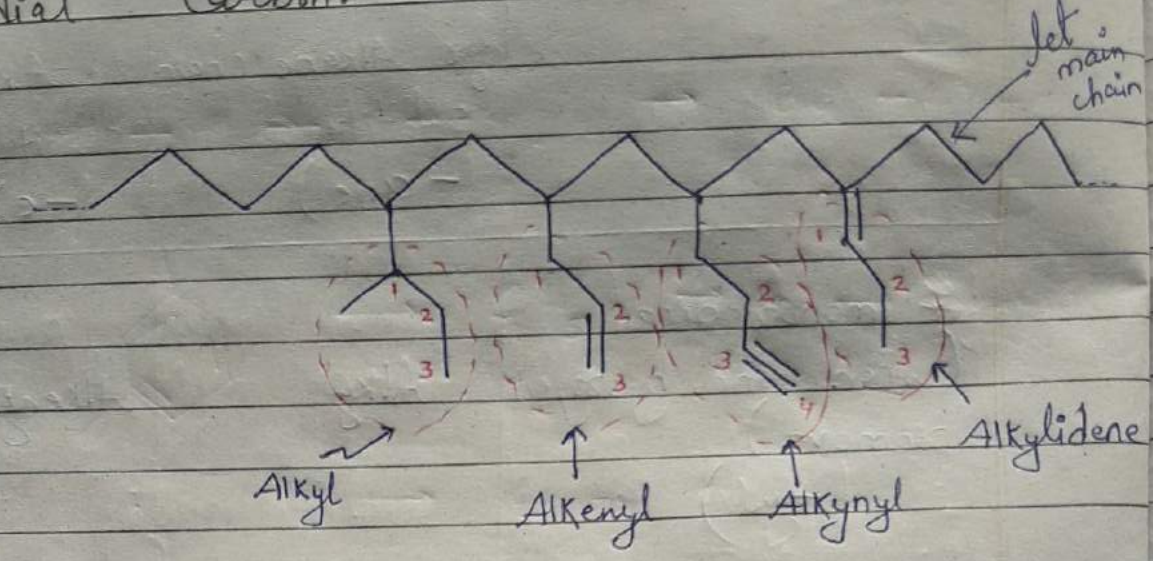
e. 2° suffix → suffix of functional group. (with position).

# I. IUPAC Nomenclature of substituents :-

| | for Alkane chain substituent (-ane) | for Alkene chain substituent (-e) | for Alkyne chain sub. (-e) |
|---|---|---|---|
| for monoradical (when one hydrogen remove) (+yl) | Alkyl | Alkenyl | Alkynyl |
| for diradical (when two hydrogen remove from same position) (+ylidene) | Alkylidene | Alkenylidene | Alkynylidene |
| for Triradicals (when three hydrogens are removed from same position) (+ylidyne) | Alkylidyne | ~~Alky~~ Alkenylidyne | Alkynylidyne |

* for substituents numbering starts from radial Carbon.

eg →




1. $CH_3$ → Methyl (Me)

2. $C_2H_3$ → Ethyl (Et)

3. $\overset{3}{C}H_3-\overset{2}{C}H_2-\overset{1}{C}H_2-$ → Propyl

4. $\overset{2}{C}H_3-\overset{1}{C}H(-CH_3)$ → 1-methylethyl (iso propyl)

5. $\overset{3}{C}H_3-\overset{2}{C}H(-CH_3)-\overset{1}{C}H_2-$ → 2-methyl propyl (iso-butyl)

6. $-F$ → fluoro

7. $-Cl$ → chloro

8. $-Br$ → Bromo

9. $-I$ → Iodo

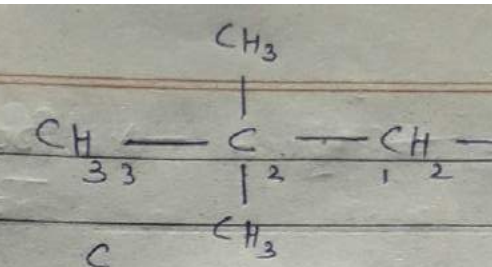
10. $-N(=O)_2$ → Nitro

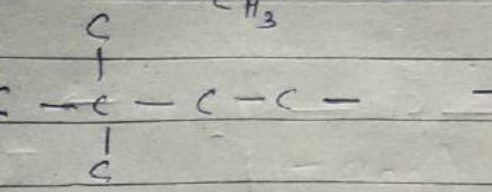
11. $-N=O$ → Nitroso

12. $CH_3-\overset{1}{C}H(-CH_3)-\overset{2}{C}H_2-\overset{3}{C}H_3$

1-methyl propyl.

Date :

13.  → 2,2-Dimethylpropyl.
~~Neo-pentyl~~ Neopentyl

14.  → 3,3-Dimethylbutyl.
Nethexyl

15. $C=C-$ → Ethenyl Vinyl

16. $\overset{3}{C}=\overset{2}{C}-\overset{1}{C}-$ → Prop-2-enyl. Allyl

17. $\overset{3}{C}-\overset{2}{C}=\overset{1}{C}-$ → Prop-1-enyl

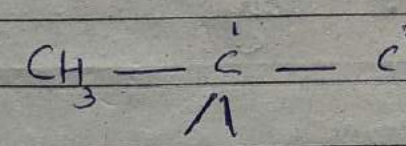
18. $\overset{3}{C}\equiv\overset{2}{C}-\overset{1}{C}-$ → Prop-2-ynyl. Propargyl

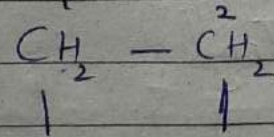
19. $CH_2<$ → Methylidene Methylene

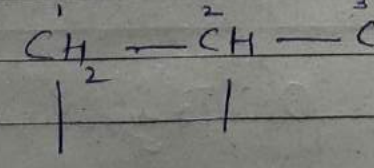
14.

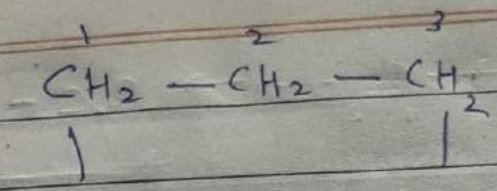
20. $\overset{2}{CH_3}-\overset{1}{CH}<$ → Ethylidene

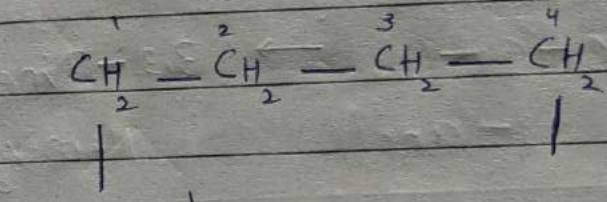
21. $\overset{3}{CH_3}-\overset{2}{CH_2}-\overset{1}{CH}<$ → Propylidene

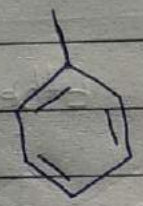
22.  → 1-methyl-ethylidene
~~Ethylidene~~

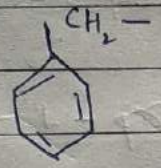
23.  → Eth-1,2-diyl
Ethylene

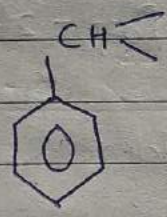
24.  → Prop-1,2-diyl
propylene

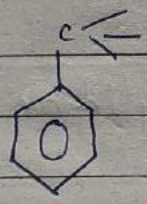
25.  → Prop-1,3-diyl
Trimethylene

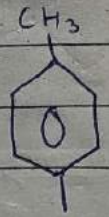
26.  → But-1,4-diyl
Tetramethylene

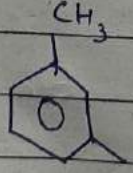
27.  ⟶ $C_6H_5$ → Ph → Phenyl

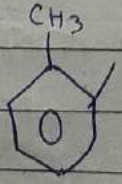
28.  → Benzyl

29.  → Benzylidene / Benzal

30.  → Benzylidyne (Benzo)

31.  → P-Tolyl

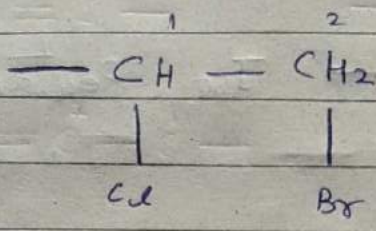
32.  → m-Tolyl

33.  → O-Tolyl

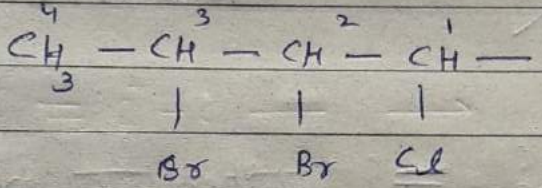
34. $-O-CH_3$ → Methoxy

35. $-O-C_2H_5$ → Ethoxy

36. $-O-Ph$ → Phenoxyl.

37. 

→ 2-Bromo-1-chloroethyl.

38. 

→ 2,3-Dibromo-1-chlorobutyl.

## Common names of Alkane & Alkyl substituents

* For Alkanes.

① n-alkane = straight chain alkane

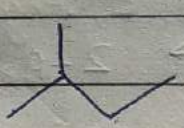
 → n-Butane

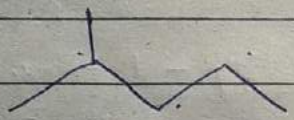
 → n-Pentane

 → n-Heptane

② Iso alkane = when one methyl group is present on straight chain

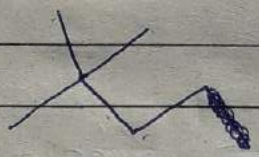
 → iso-butane

 → Iso Pentane

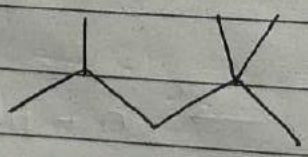
 → Iso hexane

③ Neo-alkane = when 2 methyl groups are present on straight chain.

 → Neo-pentane

 → Neo-hexane

Note :- Iso & neo prefix are applicable up to 6 Carbon Containing alkane.

 → Iso octane

→ for alkyl substituents

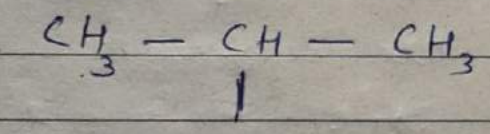
a. n-alkyl

$CH_3-CH_2-CH_2-$ n-propyl.

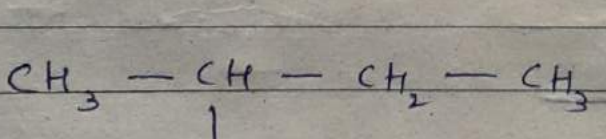
$CH_3-CH_2-CH_2-CH_2-$ n-butyl.

$CH_3-CH_2-CH_2-CH_2-CH_2-CH_2-$ n-Hexyl.

b. Sec. alkyl

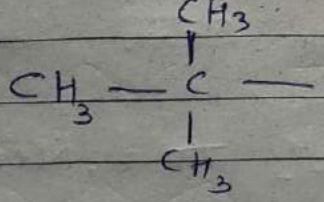



Sec alkyl / Iso propyl

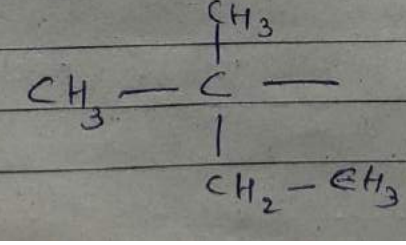



Sec. butyl

c. Tert alkyl

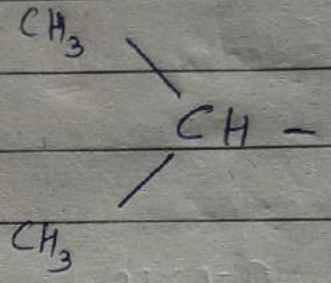



Tert butyl

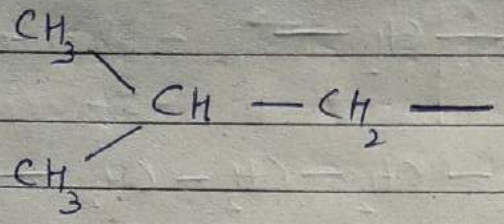



Tert. Pentyl

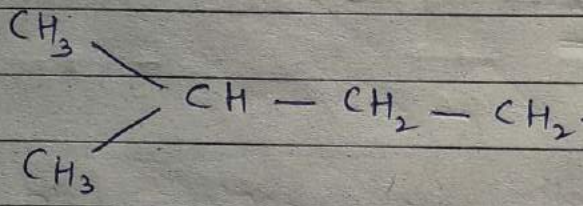
d. Iso-alkyl



Iso propyl

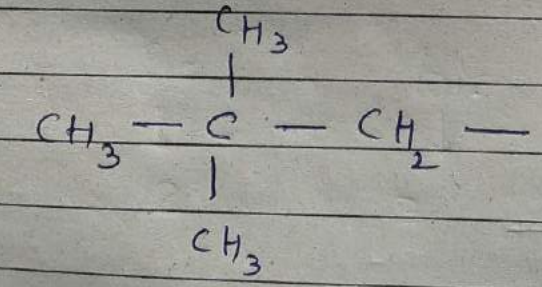


Iso butyl

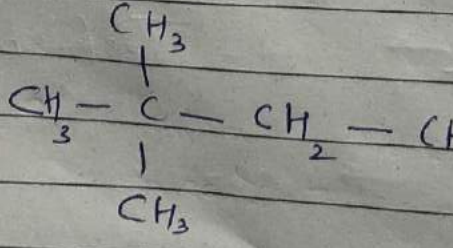


Iso hexyl

e. Neo alkyl



Neo pentyl



Neo hexyl

# III. IUPAC NOMENCLATURE Of Alkanes.

## Rule I. For selection of main/ Parent chain.

a. Select largest Continuous Carbon-Carbon chain as parent chain.

b. When more than one longest chains are present, then that chain will be parent chain which has max. no. of substituents.

c. When no. of substituents are also same then lowest set of position of substituents containing chain will be parent chain.

d. When above all conditions also same then Consides alphabetical order of substituents for main chain selection.

## Rule II. For numbering of main Chain.

a. number the main chain from that corner which has lowest set of position substituents.

b. When set of position are same from both Corners of main chain then Consider alphabetical order of name of substituents

Rule III.

a. When more than one same simple substituents are present of main chain then write following prefix before name of substituents.

Di , Tri , tetra , Penta , hexa ... etc
↓ ↓ ↓ ↓ ↓
2 3 4 5 6 ...

with position for same simple sub.

b. When more than one same complex substituents are present of main chain then write following prefix before name of substituents.

Bis , tris , tetrakis , Pentakis
↓ ↓ ↓ ↓
2 3 4 5

with position for same Complex subst.

Note :- • following are considers in alphebetical order :-

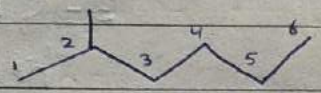
- iso
- Neo
- Di , tri , Tetra .... etc of Complex sub

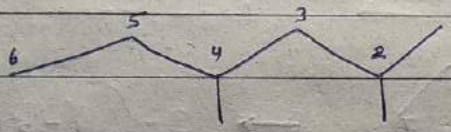
Date :

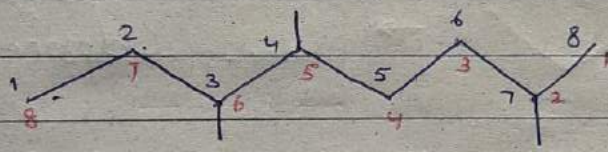
- following are not Considered in alphabetical order :-

  - Sec
  - Tert
  - Di, tri, tetra, etc of simple sub.
  - Bis, tris, tetrakis .....

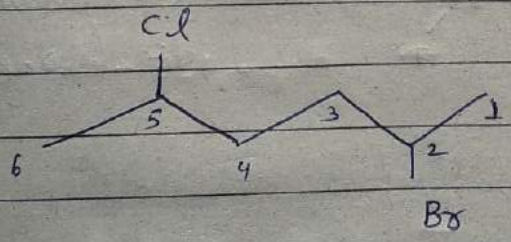
17. ★ Examples of IUPAC Name of Alkanes.

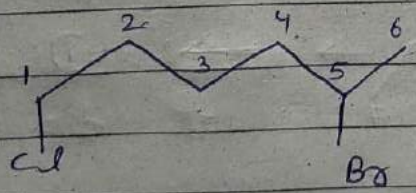
1.  → 2-Methylhexane

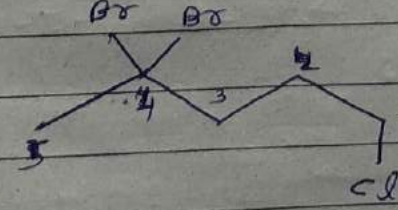
2.  → 2,4-Dimethylhexane

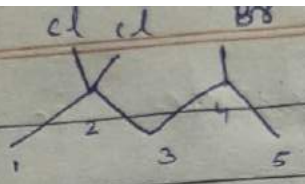
3.  → 3,4,7-Trimethyloctane ✗

   2,5,6-Trimethyloctane ✓

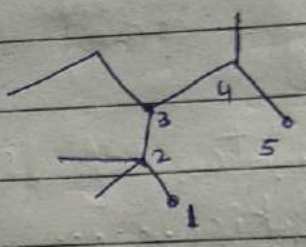
4.  → 2-Bromo-5-chloro hexane.

5.  → 5-Bromo-1-Chlorohexane

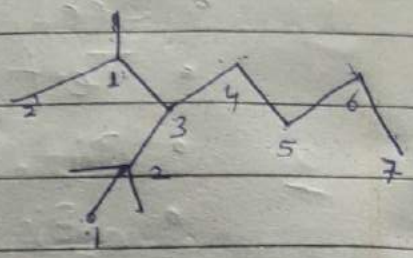
6.  → 2,2-Dibromo-1-chloro pentane.

7. 

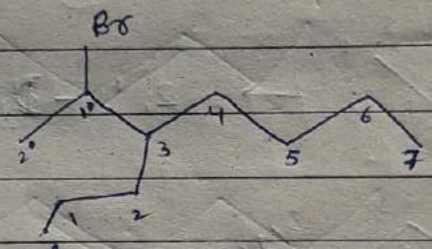
→ 4-Bromo-2,2-Chloro pentane.

8. 

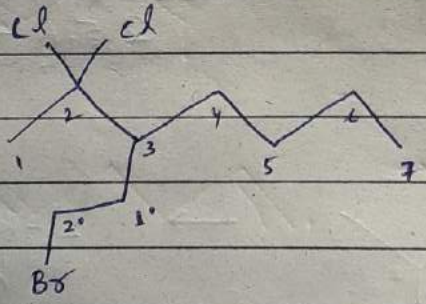
3-Ethyl-2,4-methyl pentane

9. 

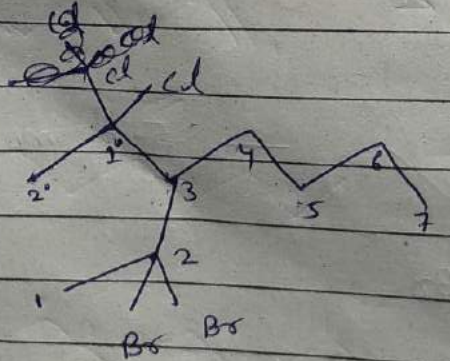
1,2-Dimethyl 3-(1-methylethyl) heptane.

10. 

→ 3-(1°-Bromoethyl)-1-Chloroheptane.

11. 

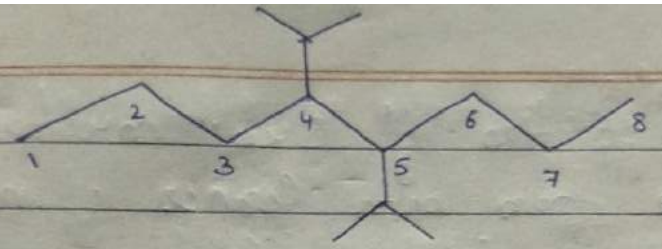
→ 3(2°-Bromoethyl)-2 dichloroheptane.

12. 

→ 2,2-Dibromo-3-(1,1-dichloroethyl) heptane

Date :

13.

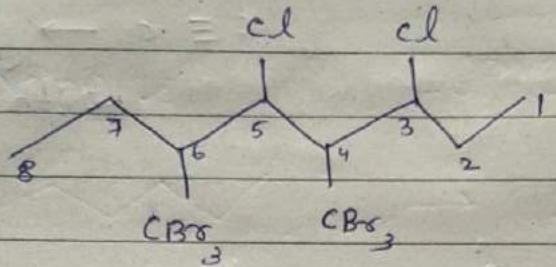



4,5-Bis (methylethyl) octane

or

4,5 - Diisopropyloctane

14.




3,5 - Dichloro - 4,6 - (bis tribromo methyl) octane.

20.

## III. IUPAC Nomenclature of Alkenes & Alkynes.

Rule 1. Select max no. of Carbon - Carbon multiple bond containing chain as parent chain According to priority order of main chain selection.

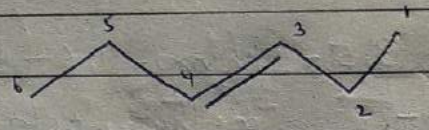
Rule 2. Give numbering from that corner of main chain which contain lowest set of position of carbon - Carbon multiple bond.

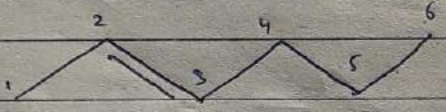
( F.G > M.B > Sub > Alph. )

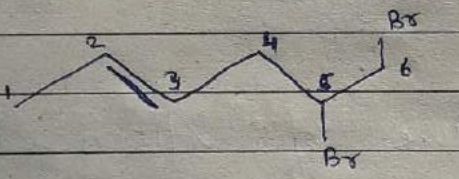
Rule 3:- When set of position of sub. multiple bond are same from both corner of main chain then Carbon - Carbon double bond is prefered on triple bond. ( C=C > C≡C )

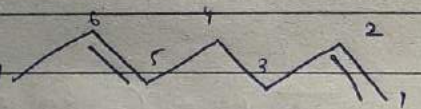
{ 1° suffix  C=C → ene
C≡C → yne } with position.

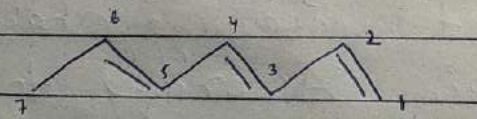
21.

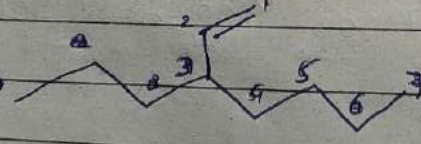
1.  → Hex-3-ene.

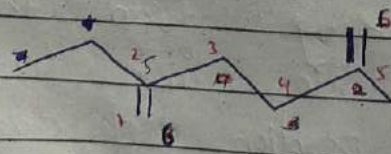
2.  → Hex-2-ene

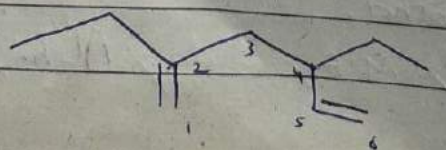
3.  → 5,6-Dibromohex-2-ene

4.  → Hepta 1,5-diene

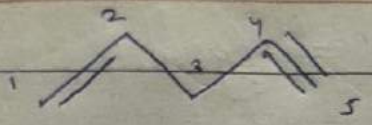
5.  → Hepta-1,3,5-triene

* 6.  → ~~Hepta 3-Ethyl~~ 3-Propylhept-1-ene

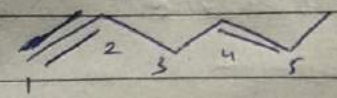
* 7.  → 2-Ethyl-5-methylhexa-1,5-diene.

8.  → 2,4-Diethylhexa-1,5-diene

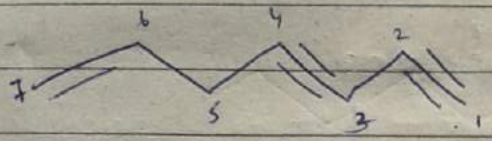
Date :

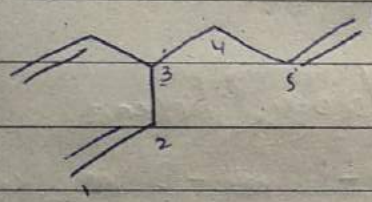
9.  → Pent-1-en-4-yne.

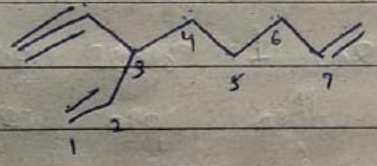
10.  → Hex-1-en-4-yne

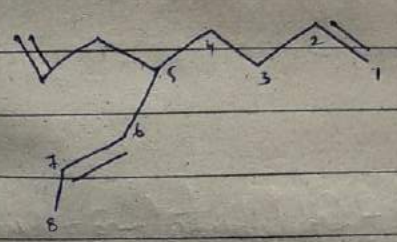
11.  →

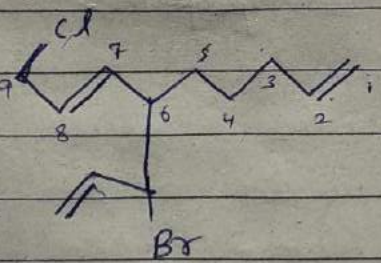
Hex-4-en-1-yne

12.  → Hept-6-ene 1,3-diyne

13.  → 3-Ethenylhexa-1,5-diene

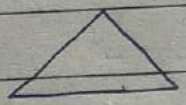
14.  → 3-Ethynylocta-1,7-diene

15.  → 5-(Prop-2-enyl)octa-1,6-diene

16.  → 6-(1-Bromoprop-2-enyl)-9-chloro nona-1,7-diene.

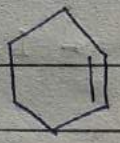
29. **IV. IUPAC Nomenclature of Aliphatic Monocyclic Compounds :-**

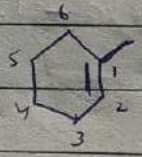
Rule 1:- Primary Prefix = Cyclo.

1.  → Cyclopropane

2. 

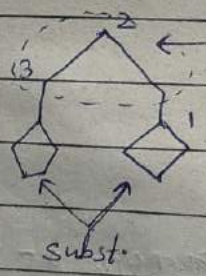
→ Chlorocyclopentane

3.  → Cyclohexene

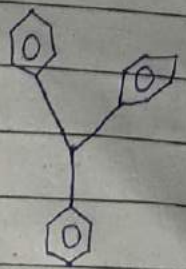
4. 

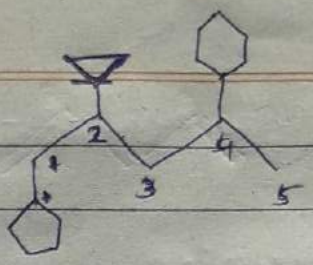
→ 1-Chlorocyclohexene

Rule 2:- When more than 1 rings are attached to open chain then open chain is treated as parent chain.

1. 

→ 1-Cyclobutyl-3-cyclopentyl propane.

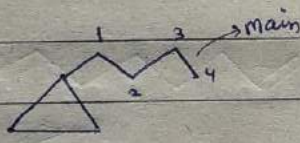
2.  → Triphenyl methane.

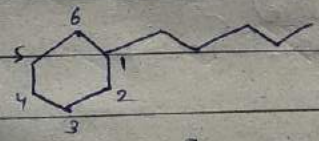
3.  → 4-Cyclohexyl-1-cyclopentyl-2-Cyclopropylpentane.

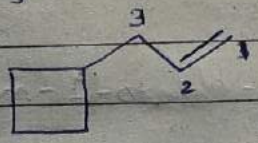
**Rule 3:-** When an open chain present on ring then select main chain according to following order.

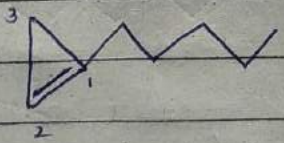
→ max. no. of functional group
→ " " Carbon-Carbon multiple bond.
→ " " Carbon atom
→ " " Substituents.

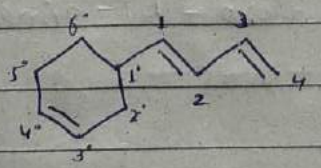
if all are same then ring will be main chain.

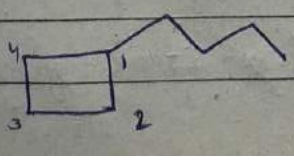
24. 1.  → 1-Cyclopropylbutane

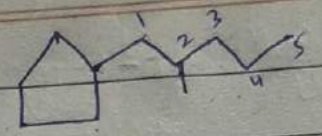
2.  → Pentylcyclohexane

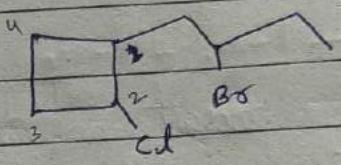
3.  → 3-Butylpropene

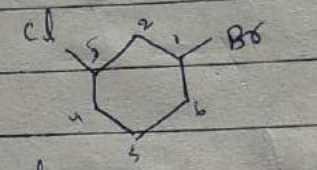
4.  → 1-Pentylcyclopropene

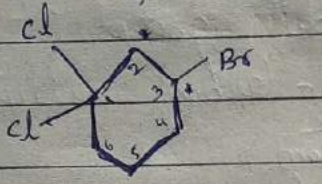
5.  → 1-(Cyclohex-3-enyl)buta-1,3-diene

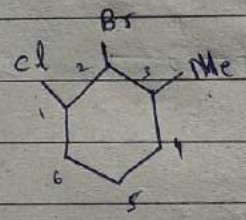
6.  → Butylcyclobutane

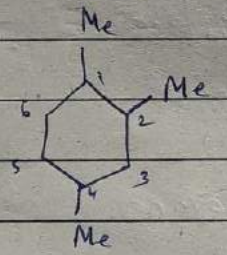
7.  → 1 - Cyclopentyl - 2 - methyl pentane

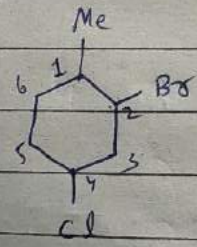
8.  → ~~Cyclopentyl~~ 2-methyl 1-(2-Bromobutyl)-2-chloro cyclo butane.

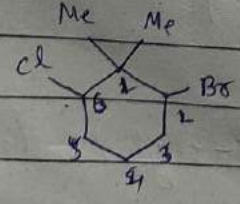
9.  → 1-Bromo-3-Chloro cyclohexane

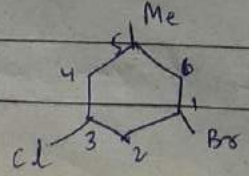
10.  → 3-Bromo-1,1-dichloro cyclohexane.

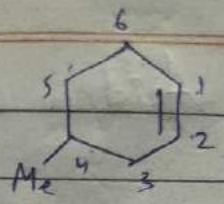
11.  → 2-Bromo-1-chloro-3-methyl cyclohexane

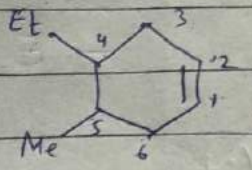
12.  → 1,2,4-Trimethyl Cyclohexane

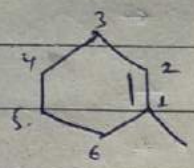
13.  → 2-Bromo-4-chloro-1-methyl cyclohexane

14.  → 2-Bromo-6-chloro-1,1-dimethyl cyclohexane

15.  → 1-Bromo-3-chloro-5-methyl cyclohexane

16.  → 4-Methylcyclohexene

17.  → 4-Ethyl-5-Methylcyclohexene.

18  → 1-Methylcyclohexene.

26. ## V. IUPAC Nomenclature of Functional Group Containing Compounds :-

Rule 1 :- Select parent chain with maximum number of principal functional groups.

Then according to priority order of main chain selection.

Rule 2 :- Give number of main chain from that corner which has lowest set of position of functional groups ( F.G > M.B > Sub > Alpha. )

Note :- when chain terminal Carbon Containing functional group is present, then numbering starts from Carbon of functional group.

1. $-\overset{O}{\overset{\|}{C}}-OH$ → Carboxylic Acid

2. $-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-$ → anhydride

3. $-\overset{O}{\overset{\|}{C}}-OR$ → ester

4. $-\overset{O}{\overset{\|}{C}}-X$ → acid halide

5. $-\overset{O}{\overset{\|}{C}}-NH_2$ → amide

6. $-CN$ → Cyanide

7. $-\overset{O}{\overset{\|}{C}}-H$ → Aldehyde.

Rule 3 :- When more than one same principal functional groups are present on main chain then use di, tri, tetra etc before suffix of functional group.

Rule 4 :- When more than one diff. functional groups are present in compound, then select principal functional group according to following order. (& other functional groups are ~~treated~~ treated as sub.)

$-COOH > -SO_3H > -\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}- > -\overset{O}{\overset{\|}{C}}-OR >$

$-\overset{O}{\overset{\|}{C}}-X > -\overset{O}{\overset{\|}{C}}-NH_2 > -CN > -NC > -\overset{O}{\overset{\|}{C}}-H >$

$-\overset{O}{\overset{\|}{C}}-R > -OH > -SH > -NH_2$

Thiolcohol amine

27.

| Functional Group | Prefix | Sufix | General Name (Alkyl-F.G) |
|---|---|---|---|
| 1. $-\overset{O}{\overset{\|}{C}}-OH$ | Carboxy | oic Acid | Alkanoic acid |
| 2. $-\overset{O}{\overset{\|}{S}}(=O)_2-OH$ | Sulpho | sulphonic Acid | Alkanesulphonic acid |
| 3. $-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-$ | — | oic anhydride | Alkanoic anhydride |
| 4. $-\overset{O}{\overset{\|}{C}}-OR$ | Alkoxy Carbonyl<br>Alkanoyl oxy | oate | Alkyl alkanoate |
| 5. $-\overset{O}{\overset{\|}{C}}-X$ | Halocarbonyl | oyl halide | Alkanoyl halide |
| 6. $-\overset{O}{\overset{\|}{C}}-NH_2$ | Carbamoyl | amide | Alkanamide |
| 7. $-CN$ | Cyano | nitrile | Alkanenitrile |
| 8. $-N\equiv C$ | Isocyano | isonitrile | Alkanisonitrile |
| 9. $-\overset{O}{\overset{\|}{C}}-H$ | formyl & oxo | al | Alkanal |
| 10. $-\overset{O}{\overset{\|}{C}}-R$ | oxo or keto | one | Alkanone |

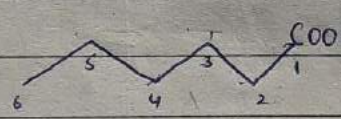
| | | | |
|---|---|---|---|
| 11. $-OH$ | Hydroxy | ol | Alkanol |
| 12. $-SH$ | Mercapto | thiol | Alkanethiol |
| 13. $-NH_2$ | Amino | amine | Alkamine |

28. Elementary Examples of Functional Group Containing Compounds :-

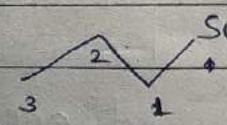
(I) $-COOH$ Suffix → oic acid

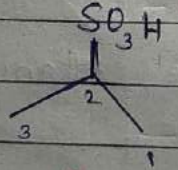
1. $H-\overset{1}{C}OOH$ → Methanoic Acid

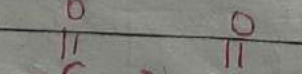
2. $\overset{2}{C}H_3-\overset{1}{C}OOH$ → Ethanoic Acid

3.  → Hexanoic Acid

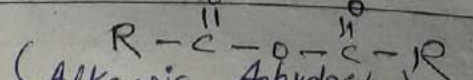
(II) $-SO_3H$ Suffix → sulphonic acid

1.  → Propane-1-sulphonic acid.

2.  → Propane-2-sulphonic acid

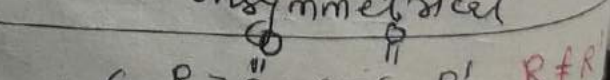
(III)  Anhydride Suffix → oic anhydride

Symmetrical

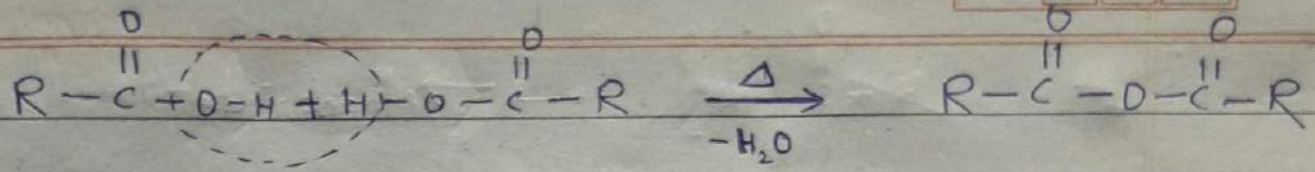


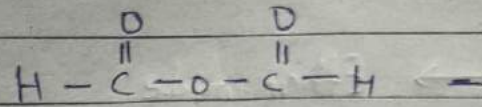
(Alkanoic Anhydride)

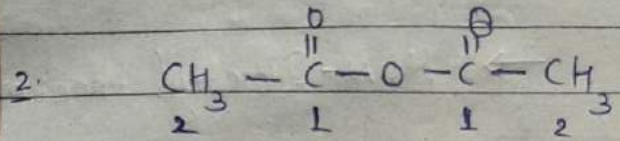
Unsymmetrical

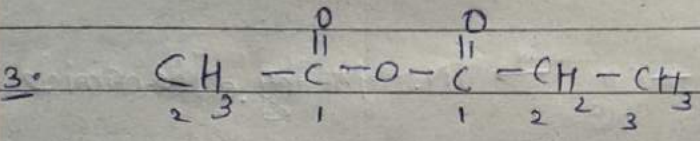


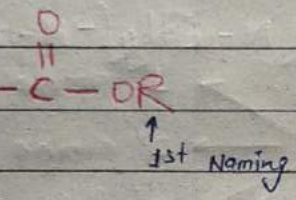
(Alkanoic Alkanoic Anhydride)

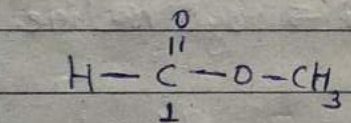


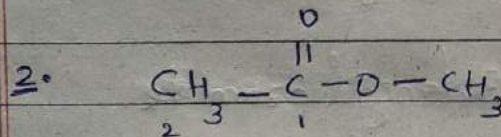
1.  → Methanoic anhydride (unstable)

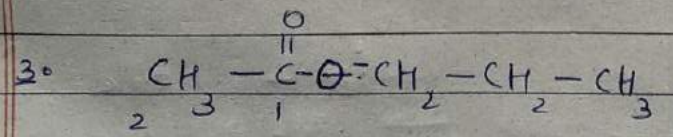
2.  → Ethanoic anhydride

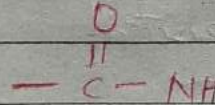
3.  → Ethanoic propanoic anhydride

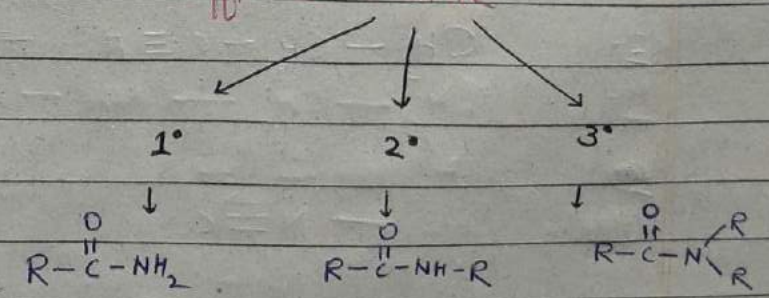
(IV)  suffix → oate

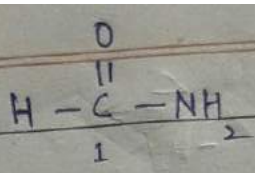
1.  → Methyl methanoate

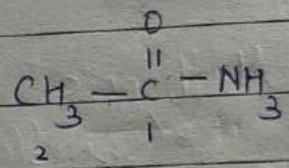
2.  → Methyl ethanoate

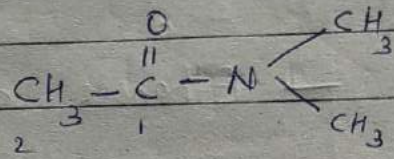
3.  → Propyl ethanoate

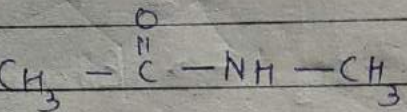
(V)  suffix → amide

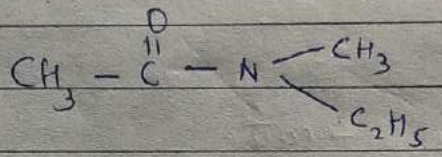


1.  → Methanamide

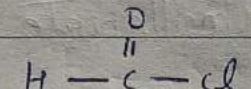
2.  → Ethanamide

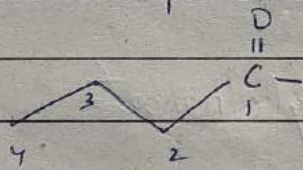
3.  → N, N- Dimethylethanamide

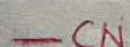
4.  → N- Methylethanamide

5.  → N- Ethyl-N-methylethanamide

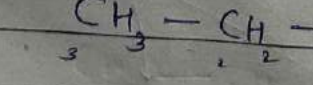
(VI)  suffix → oyl halide

1.  → Methanoyl Chloride (unstable)

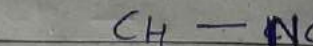
2.  → Butanoyl bromide

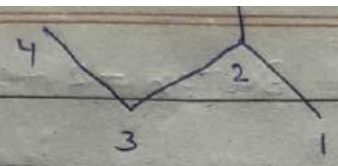
VII  suffix → nitrile

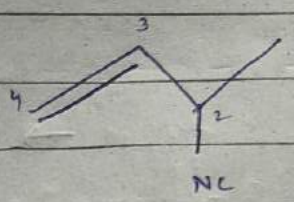
1.  → Methanitril

2.  → Propanenitrile

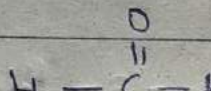
VIII.  suffix → isonitrile

1.  → Methanisonitrile

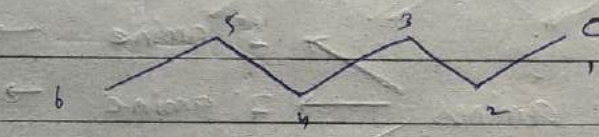
2.  → Butan-2-isonitrile

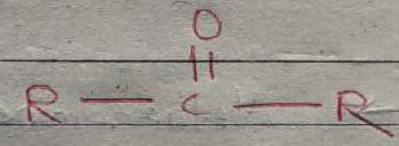
3.  → But-3-en-2-isonitrile.

IX.  suffix → al.

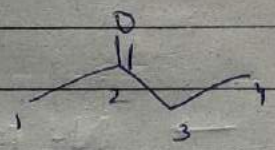
1.  → Methanal

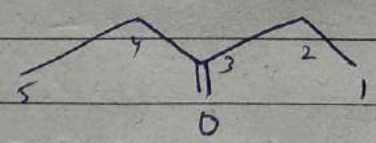
or H-CHO

2.  → Hexanal.

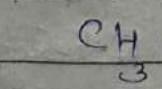
X.  suffix → one

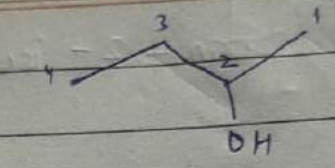
1.  → Propanone

2.  → Butanone

3.  → Pentan-3-one

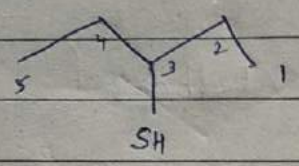
XI  suffix → ol.

1.  → Methanol

2.  → Butan-2-ol

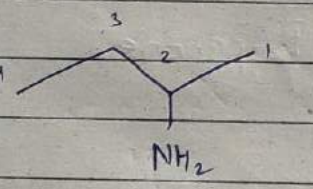
XII. —SH suffix → thiol

1. $CH_3 - SH$ → Methanthiol

2.  → Pentan-3-thiol.

XIII. $-NH_2$ amines → 1° amine → $R-NH_2$
→ 2° amine → $R-NH_2-R$
→ 3° amine → $R-NH_2-R$ (R)

1. $CH_3 - NH_2$ → Methanamine

2.  → Butan-2-amine

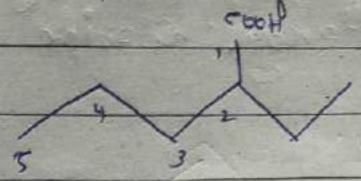
3. $CH_3 - NH - CH_3$ → N-methylmethanamine
(subst.)

4. $CH_3 - CH_2 - NH - CH_3$ → N-methylethanamine

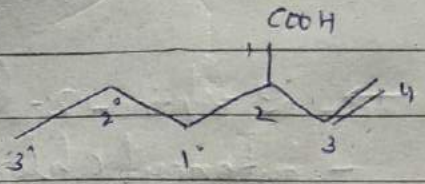
5. $CH_3 - CH_2 - CH_2 - N(CH_3)(C_2H_5)$ → N-Ethyl-N-methyl Propan-1-amine
~~N-methyl-N-ethyl~~

29° → Examples of funtional group. { Advanced }

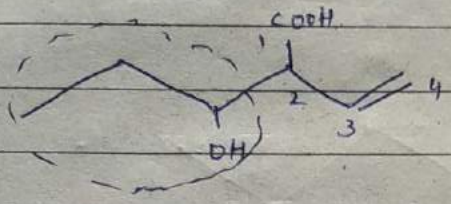
30°

1. 

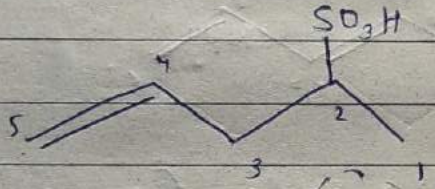
→ 2-Ethylpentanoic acid

2. 

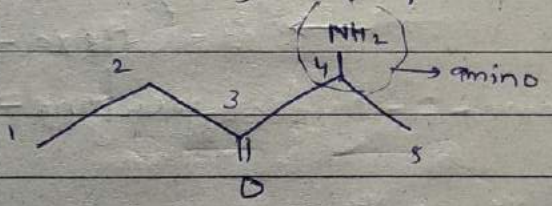
→ 2-Propylbut-3-enoic acid

3. 

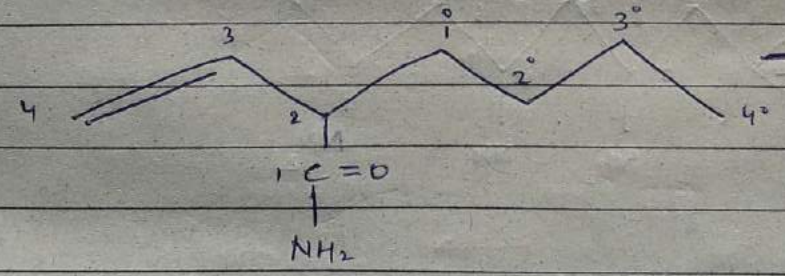
→ 2-(1-Hydroxypropyl) but-3-enoic acid

4. 

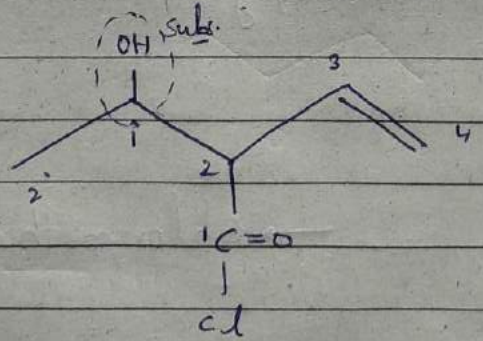
→ Pent-4-ene-2-sulphonic acid

5. 

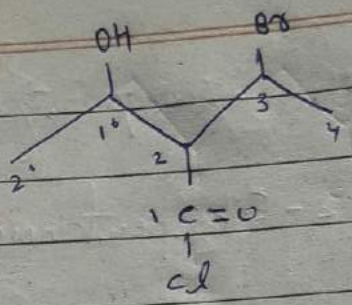
→ 4-Aminopent-1-en-3-one

6. 

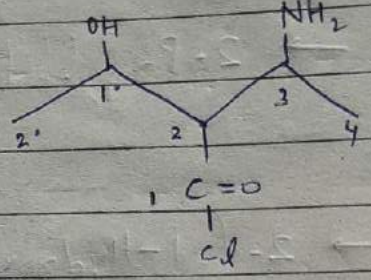
→ 2-Butylbut-3-enamide

7. 

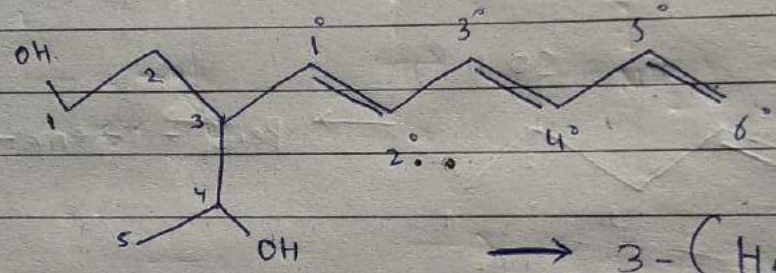
→ 2-(1-Hydroxyethyl)-but-3-enoyl chloride

8. 

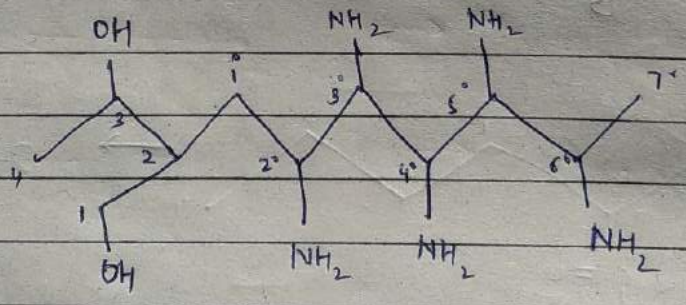

3-Bromo-2-(1-hydroxyethyl) butanoylchloride

9. 


→ 3-Amino-2-(1-hydroxyethyl) - butanoyl chloride.

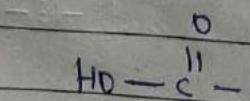
10. 


→ 3-(Hexa-1,3,5-trienyl) pentane-1,4-diol.

11. 


2-(2,3,4,5,6-Pentaaminoheptyl) butane-1,3-diol.

30

12. 


→ Chloromethanoic acid

H-O-COCl

13. $HO-CHO$ } ⟶ Methanoic acid

or $HO-\overset{\overset{O}{\|}}{C}-H$ }

14. $NH_2-CHO$ } ⟶ Methanamide

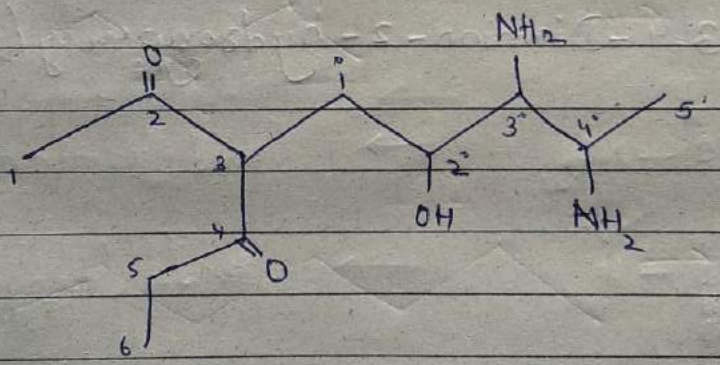
$NH_2-\overset{\overset{O}{\|}}{C}-H$ }

15. $HO-CONH_2$ ⟶ Aminomethanoic acid

16. $NH_2-\overset{\overset{O}{\|}}{C}-NH_2$ } ⟶ Aminomethanamide.

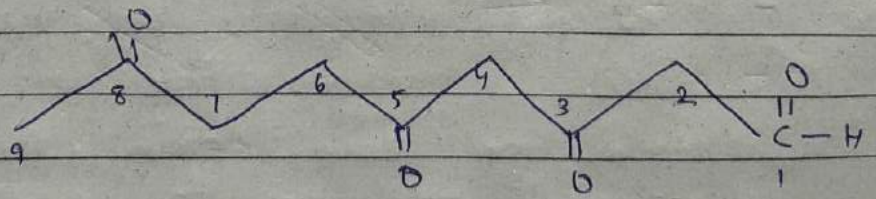
(Urea)

17.



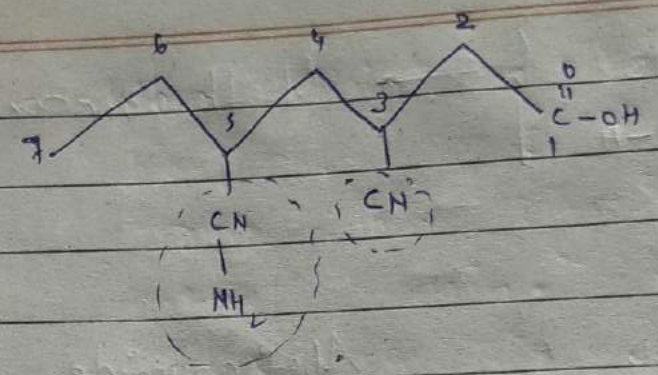

3-(3,4-Diamino-2-hydroxypentyl)-hexane-2,4-dione.

18.



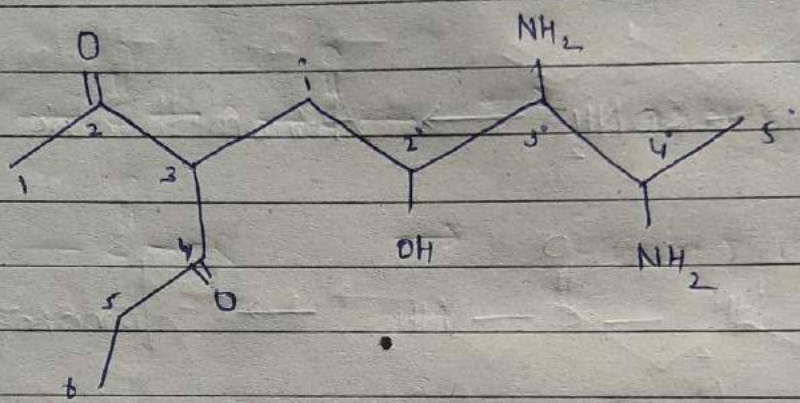

3,5,8-Trioxononanal.

A.




5-Carbamayl-3-Cyano-heptanoic acid

20.




3-(3,4-Diamino-2-hydroxypentyl)-hexan-2,4-dione

21.

Note :- When chain terminating Carbon Containing functional group directly bounded with ring then ring is treated as parent chain. & following special suffix use for functional group.

1. $-\overset{\overset{O}{\|}}{C}-OH$ → Carboxylic acid

2. $-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-$ → Carboxylic anhydride.

3. $-\overset{\overset{O}{\|}}{C}-OR$ → Carboxylate

4. $-\overset{\overset{O}{\|}}{C}-Cl$ → Carbonyl chloride

5. $-\overset{\overset{O}{\|}}{C}-NH_2$ → Carboxamide.

6. $-\overset{\overset{O}{\|}}{C}-H$ → Carbaldehyde

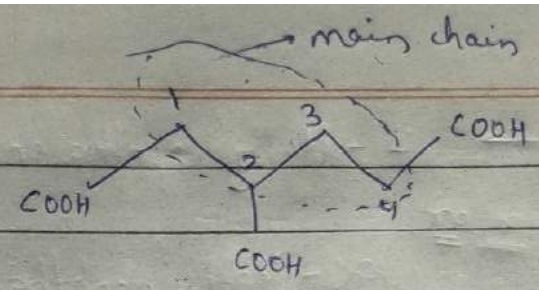
7. $-CN$ → Carbonitrile.



Note :- When more than two chain terminating Carbon Containing F.G are directly bonded to open chain, then open chain is treated as parent chain & special suffix are used for functional group.

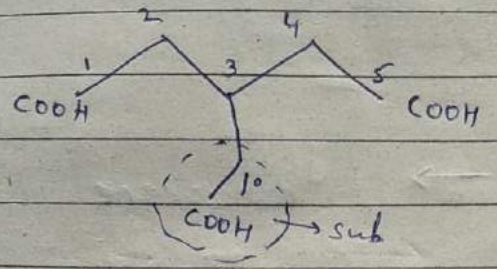
Date :

1.




Butane - 1, 2, 4 - tricarboxylic acid

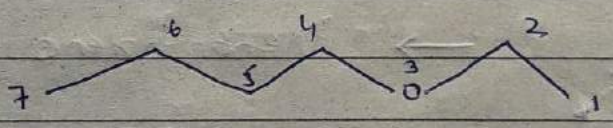
2.




3 - (carboxymethyl) pentanedioic acid.

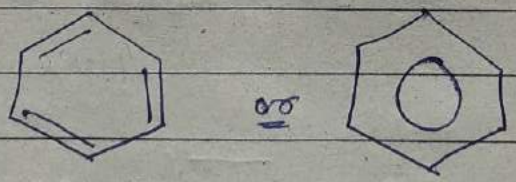
## Replacement Nomenclature for heteroatom Containing chains :-

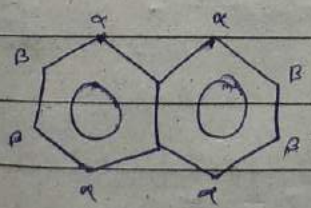
1.

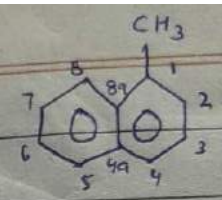


3 - Oxaheptane.

## 25. IUPAC Nomenclature of Aromatic Compounds :-

1.  → Benzene
{not Cyclohexa-1,3,5-triene}

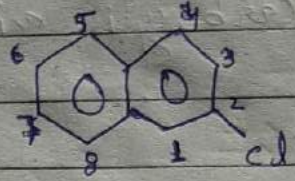
2.  → Naphthalene

•


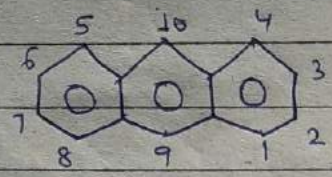
→ 1 - Methylnaphthalene

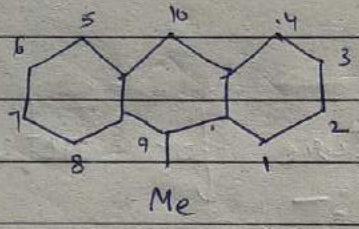
or

α - Methylnaphthalene

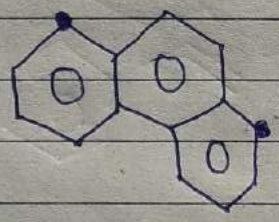
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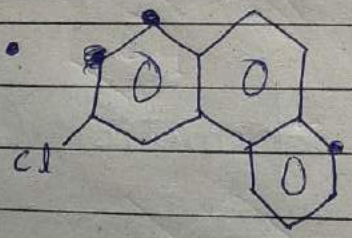
→ 2 - Chloronaphthalene

3.

→ Anthracene

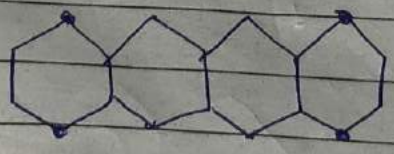
4. •


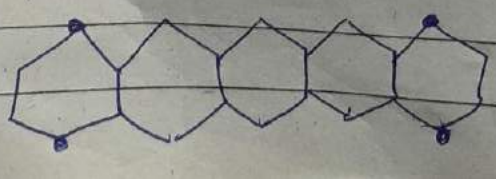
→ 9 - Methylanthracene

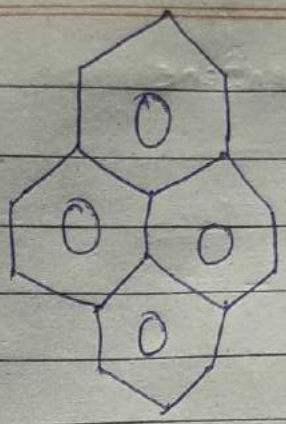
4.

→ Phenanthrene

•


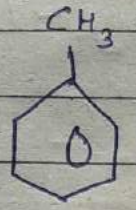
→ 3 - Chlorophenanthrene

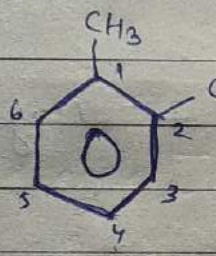
5.

→ Tetracene

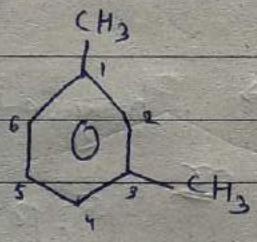
6.

→ Pentacene

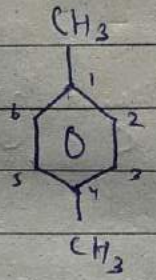
7.  → Pyrene

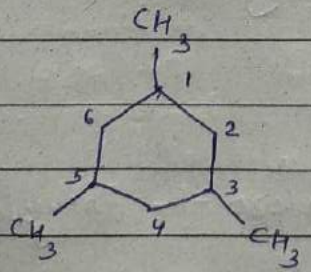
Examples of substituted aromatic Comp-

1.  → Methylbenzene (Toliene)

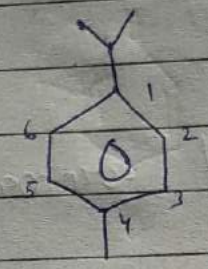
2.  → 1,2-Dimethylbenzene
or Ortho Dimethylbenzene
or O-Xylene

3.  → Metadimethylbenzene
or Meta Xylene

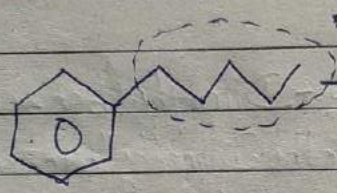
4.  → Paradimethylbenzene
or P-Xylene

5.  → 1,3,5-Trimethylbenzene
or Symetrical trimethylbenzene
or
Mesitylene

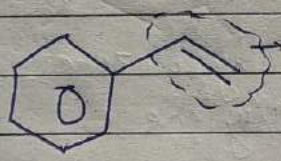
6.  → Isopropylbenzene
or
Cumene

7.  → Cymene or
1-Isopropyl-4-methylbenzene

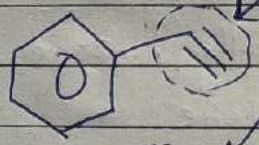
Note :-

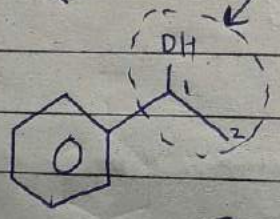
8. 

→ 1-Phenylpentane

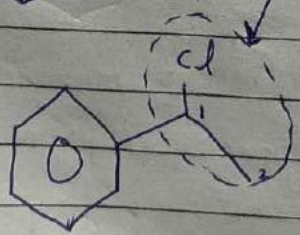
9. 

→ Phenylethene
or
Styrene

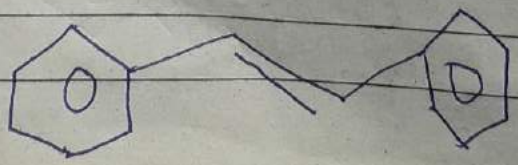
10.  → Phenylethyne

11. 

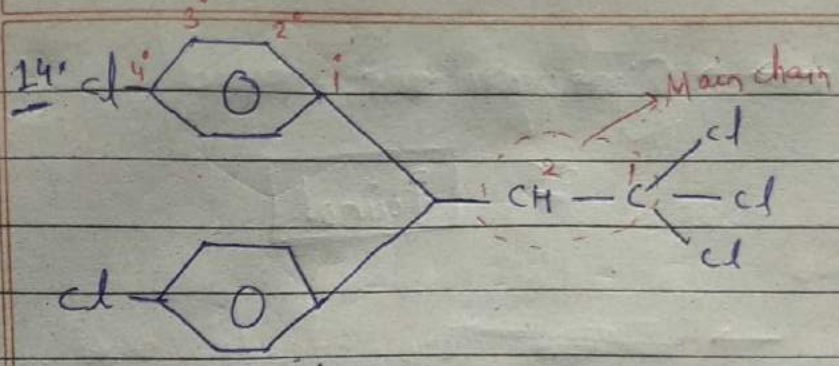
→ 1-Phenylethan-1-ol

12. 

→ 1-Chloro-1-Phenylethane

13.  → 1,2-Diphenylethene
or
(Stilbene)

Date :

14. 


1,1,1 – Trichloro – 2,2 – bis (4 – chlorophenyl) ethane

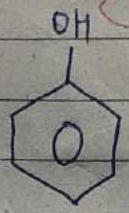
or

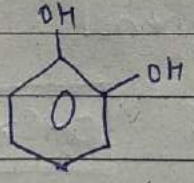
P, P' Dichloro Diphenyl Trichloroethane

(D.D.T)

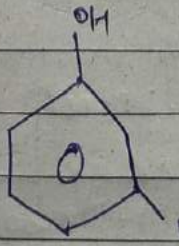
36. Functional group Containing aromatic Compounds:-

1. – OH group Containing

1.  → Phenol (carbolic acid)

2.  → Benzene – 1,2 – diol

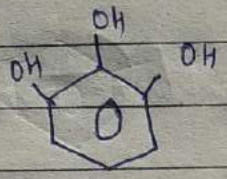
or Catechol

3.  → Resorcinol.

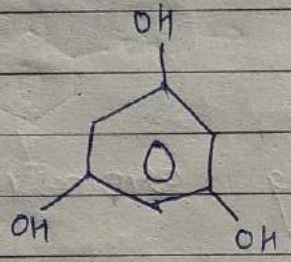
4.  → Hydroquinone

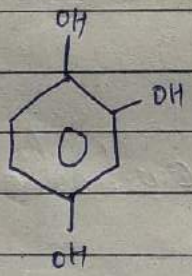
or Quinol

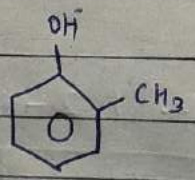
5.  → Quinone

6.  → Benzene – 1, 2, 3 – triol

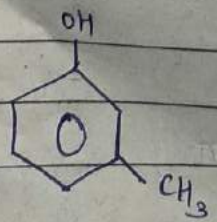
or Pyrogallol

7.  → Phloroglucinol

8.  → Hydroxyquinol

9.  → 2 – Methylphenol

or O – Cresol

10.  → 3 – Methylphenol

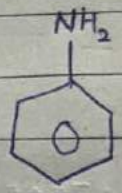
or m – Cresol

Date :

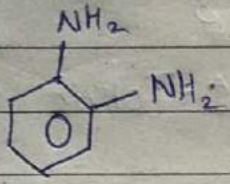


P-Cresol

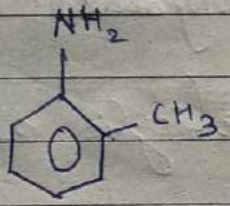
2. $-NH_2$ group Containing



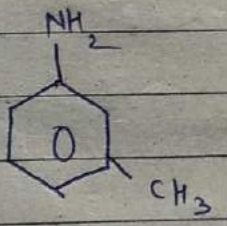
→ Aniline & Benzenamine



→ O-Phenylenediamine

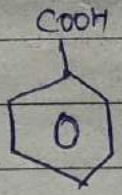


→ 2-Methylaniline or O-Toluidine



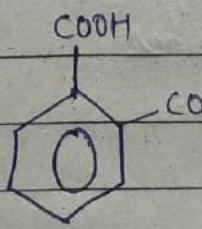
→ m-Toluidine

3. $-COOH$ group Containing



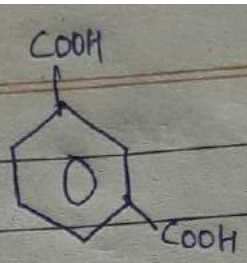
→ Benzenecarboxylic acid

or Benzoic Acid

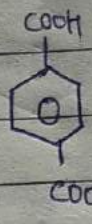


→ Benzene-1,2-dicarboxylic acid

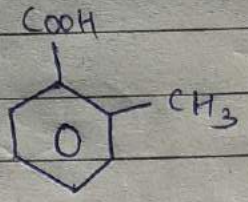
or Phthalic acid.



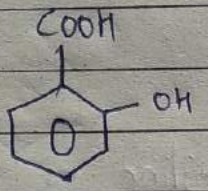

→ Isophthalic acid




→ Terephthalic acid



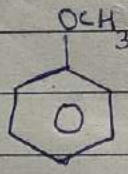

→ 2-Methylbenzene Carboxylic acid
or O-Toluic acid



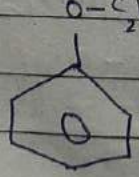

→ O-Hydroxybenzoic acid
or Salicylic acid

37

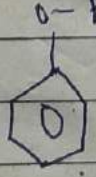
3. E-ther




→ Methoxybenzene
or Anisole
or Methylphenylether

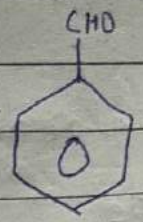



→ Ethoxybenzene
or Phenetole
or ethylphenylether

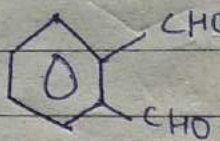



→ Phenoxy benzene
or Diphenyl-ether

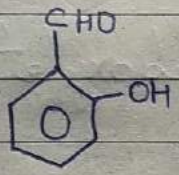
4. =CHO group containing




→ Benzonecarbaldehyde
or
Benzaldehyde

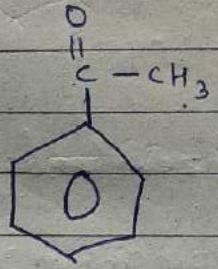



→ Benzone - 1,2 - dicarbaldehyde
or
Phthalaldehyde.

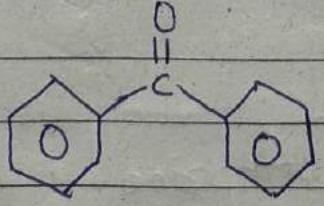



2 - Hydroxybenzenecarbaldehyde
or
Salicaldehyde.

5. Ketonic group contains

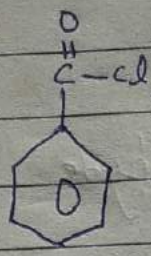



→ 1-Phenylethan-1-one
or Acetophenone

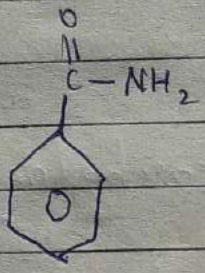



→ Diphenylmethane
or Diphenyl ketone
or Benzophenone

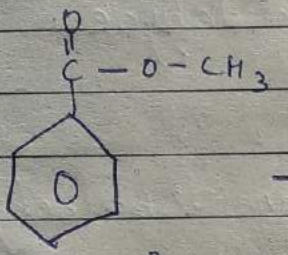
## 6. Derivatives of Carboxylic acid



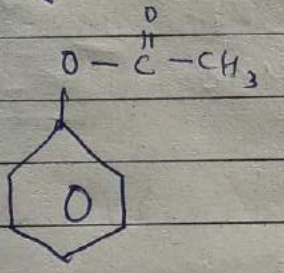
→ Benzene Carbonyl chloride
or Benzyl chloride



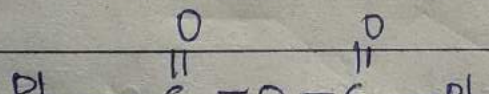
→ Benzenecarboxamide
or Benzamide



→ Methyl benzene carboxylate
or Methyl Benzoate

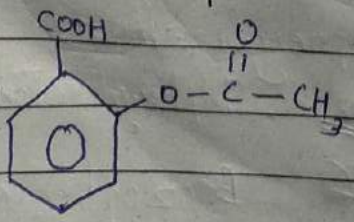


→ Phenylethanoate
or Phenylacetate

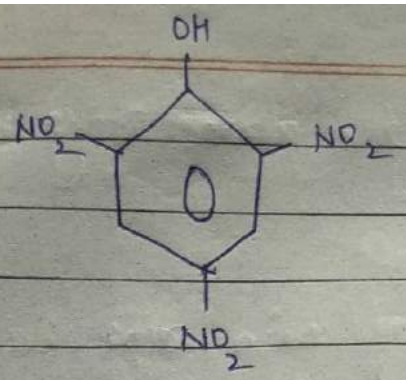


→ Benzene Carboxylic anhydride
or Benzoinanhydride.

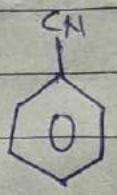
Other examples :-



→ Aspirine
or Acetyl salicylic acid



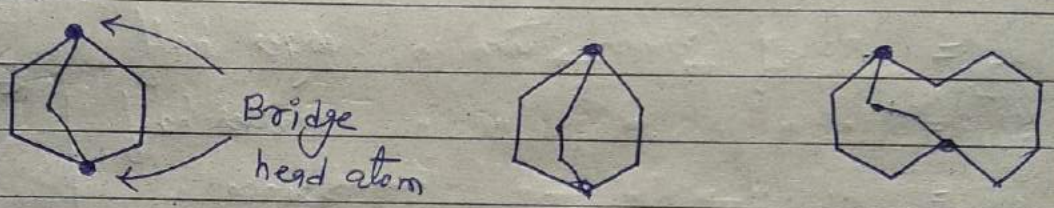

→ Picric acid




→ Benzene carbonitrile

or Benzonitrile

## 38. IUPAC Nomenclature of Bicyclic Compound.




Rule 1:- Select largest ring (with max. no of Carbon atoms) as main ring

{ max. C atoms > max. FG > Max m.B > max. sub > lowest locant > alphabetical order }

Rule 2:- Numbering starts from bridge head atom {always}.

{ lowest position (F.G > M.b > sub) > alp. order }

* Sequence of numbering main ring.

a. Bridge head atom
b. larger half part of main ring
c. Smaller " " " " "
d. in bridge

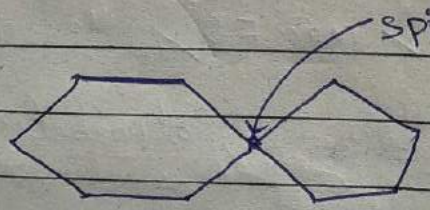
Rule: "Bicyclo" = 'i' prefix

Sub + bicyclo (l, s, b) + word root + (ane/ene/yne) + (suffix of F.G)

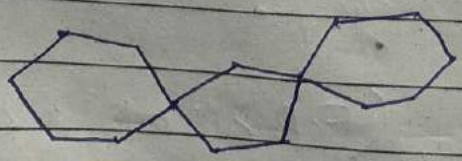
l = no. of Carbon atom in larger half
s = " " " " " " smaller half
b = " " " " " " bridge

} Excluding bridge head atoms.

# IUPAC Nomenclature of Spiro Compounds.




spiro compound.



→ bispiro

Rule 1:- Numbering starts from adjacent position of spiro carbon of smaller ring.

Rule 2:- Sequence of numbering (in small ring → larger ring).

Rule 3:- Sequence of Name

$$\text{sub} + \underset{2^\circ}{\text{Spiro}}\ (s,l) + \underset{1^\circ}{\text{word root}} + (\text{ane/ene/yne}) + \underset{\text{suffix of F.G}}{2^\circ}$$

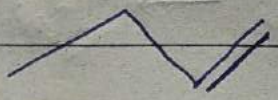
s = no. of Carbon atom in smaller ring except spiro carbon
l = " " " " " larger " " " " .

# COMMON NAMES Of Organic Compounds :-

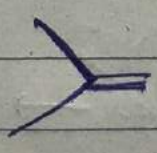
## (A) → Alkenes.

1. $CH_2 = CH_2$ → Ethylene

2.  → Propylene

3.  → 1-Butylene

4.  → 2-Butylene

5.  → Isobutylene

(B) → Alkyne.

1. $HC \equiv CH$ → Acetylene

2. $CH_3 - C \equiv CH$ → Methylacetylene

3. $CH_3 - C \equiv C - CH_3$ → Dimethylacetylene

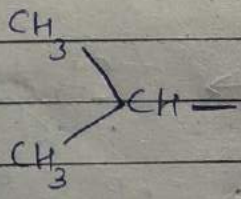
4. $Ph - C \equiv CH$ → Phenylacetylene.

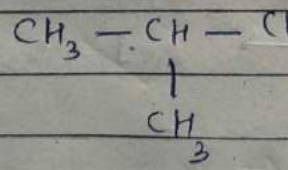
(C) $R - X$ Alkylhalide.

1. $CH_3 - Cl$ → Methyl chloride

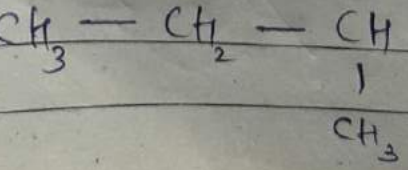
2. $CH_3 - CH_2 - Cl$ → Ethylchloride

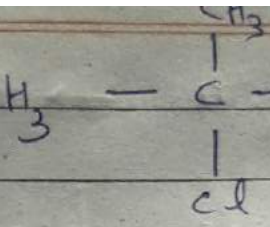
3. $CH_3 - CH_2 - Cl$ → n-propylchloride

4.  → Isopropylchloride

5.  → Isobutylchloride

6. $CH_3 - CH_2 - CH_2 - CH_2 - Cl$ →

7. 

8.  →

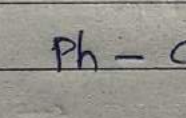
9. $CH_2 = CH - Cl$ → Vinylchloride

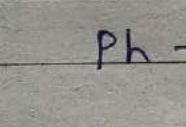
10. $CH_2 = CH - CH_2 - Cl$ → Allyl chloride

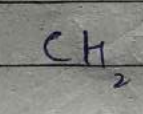
11. $Ph - Cl$ → Phenylchloride

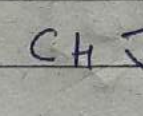
12. $Ph - CH_2 - Cl$ → Benzylchloride

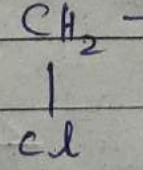
13. $CH \equiv CH - CH_2 - Cl$ → Propargylchloride

14.  → Benzochloride

15.  → Benzalchloride

16.  → Methylenechloride

17.  → Methylidyne chloride (Chloroform)

18.  → Ethylene dichloride